is the one's complement—it flips all the bits. The [] operator can be used to read (but not write) the bit at the given offset in the binary representation of the integer.

```
0b0111 & 0b0100 # => 4
0b0111 | 0b0100 # => 7
0b0111 ^ 0b0100 # => 3
~0b0110         # => -7
1.even?         # => false
2.odd?          # => false
```

The method gcd takes an integer argument and returns the greatest common denominator of the two integers. The lcm method returns the least common multiple of the two integers. The gcdlcm method returns a two element array that returns the greatest common denominator and the least common multiple of the two integers.

```
10.gcd(15)    # => 5
10.lcm(15)    # => 30
10.gcdlcm(15) # => [5, 30]
```

Integers can be used in ranges, which means integers implement next to return the next integer going upward and pred to return the previous integer going downward:

Integers add the query methods even? and odd?, which return true or false based on the value of the integer.

The chr method converts the integer to a single character string represented by that integer in an particular encoding, the default is UTF-8, but you can pass any encoding as an optional argument.

The Integer class also includes some base-transitioning methods. The digits method returns an array of all the individual digits in the integer (but reversed, the ones digit is the first digit of the array). An optional base argument puts the digits in whatever base you want as long as the base is greater than two. The default is base ten.

You can use an integer to drive a loop using the times method, which takes a block and calls the block with every integer value from 0 to the receiving value minus one. If you want to do this starting at the integer rather than ending at it, the upto method takes a required limit that is higher and calls the block for each integer from the receiver up to the limit, inclusive. To go the other direction, you can use the downto method, which takes a required limit argument that is lower and calls the block for each integer value from the receiver value down to the value of the limit, inclusive. Without a block, all these methods return an Enumerator.

Integers are convertible to other objects with to_d for BigDecimal, to_f for Float, to_i and to_int (which just returns the integer), to_r for Rational, and to_s for String.

Float Class

The Float class implements several methods for performance purposes but doesn't add a whole lot to the functionality of Numeric. Float defines constants, some of which are:

- DIG—Machine dependent; the number of significant digits in a double-precision floating point. (DIG for digits, not for digging.) Usually 15 on most current Ruby implementations.
- EPSILON—Machine dependent, the smallest difference between two numbers that can be represented. The documentation says it usually defaults to 2.2204460492503131e-16.